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School of Engineering – Computer Engineering Program

Data Analysis Programming – Semester 2026-II

Workshop No. 1

Hazard Definition, Source Validation, and Data Contract

Team 8 – Tsunami Monitoring

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Contents

1 Hazard Definition **2**
- 1.1 Phenomenon and Severity Measure . . . 2
- 1.2 Why Monitoring Matters . . . 2
- 1.3 Geographic and Temporal Scope . . . 3
- 1.4 Meaning of "Current Situation" . . . 3

2 Historical Sources **4**
- 2.1 NOAA/NCEI HazEL – Global Historical Tsunami Database . . . 4
- 2.2 USGS FDSN Event Web Service . . . 5

3 Incremental Sources **8**
- 3.1 USGS Real-Time GeoJSON Feed . . . 8
- 3.2 tsunami.gov Warning Centers . . . 9
- 3.3 Empirical Update Behavior . . . 9
- 3.4 Revised or Deleted Records and Watermark Implications . . . 10

4 Source Profiling and Data Quality **11**
- 4.1 Profiling Summary . . . 11
- 4.2 Data-Quality Problems . . . 12
- 4.3 Cross-Source Overlap . . . 13

5 Canonical Mapping **14**
- 5.1 Field Mapping to the Canonical `event` . . . 14
- 5.2 Domain Tables . . . 15
- 5.3 Unit Conversions and Rules . . . 15
- 5.4 Draft ER Diagram . . . 15

6 Ingestion Contract Draft **15**
- 6.1 Correctness After a Failed Run . . . 15

7 Project Definition **15**
- 7.1 Project Overview . . . 15
- 7.2 Functional User . . . 16
- 7.3 User Stories . . . 16
- 7.4 Indicator Proposal . . . 16
- 7.5 Risks and Mitigation . . . 16
- 7.6 Architecture Overview . . . 16

Conclusions and Open Questions **17**

A Real Response Snippets **17**
- A.1 NOAA HazEL – tsunami events . . . 17
- A.2 USGS FDSN – earthquake catalog . . . 17
- A.3 USGS real-time feed . . . 18
- A.4 tsunami.gov – PTWC Atom feed . . . 18

Executive Summary

[TODO: Five to eight lines: hazard, scope, the four sources, main findings of the profiling, and the proposed indicator.]

1 Hazard Definition

1.1 Phenomenon and Severity Measure

A tsunami is a series of long ocean waves generated by the sudden displacement of a large volume of water. Most tsunamis are caused by large undersea earthquakes, but landslides, volcanic eruptions and meteorological events can also generate them. In our HazEL sample, 36 of 221 events (16%) have a non-earthquake cause (landslide 17, meteorological 9, volcano 6, unknown 2, among others).

The canonical severity of the system is the earthquake magnitude, stored together with its type (for example mww, mwc, mb or Ml), because magnitude types differ across sources. Depth, in kilometers, is kept as a hazard-specific attribute. Since no single measure describes tsunami severity, the system combines the variables in Table 1.

Measure	Unit	Source	Meaning
Earthquake magnitude (`mag`, `eqMagnitude`)	Dimensionless; the magnitude type is stored	USGS, HazEL, tsunami.gov	Size of the earthquake that may generate a tsunami
Earthquake depth (`depth`, `eqDepth`)	km (inferred for HazEL)	USGS, HazEL, tsunami.gov	Shallow events are more likely to displace the seafloor
Maximum water height (`maxWaterHeight`)	m	HazEL	Observed tsunami height (historical only)
Event validity (`eventValidity`)	Ordinal scale from -1 to 4	HazEL	Confidence that the event was a real tsunami, from erroneous or doubtful entry to definite tsunami
Official alert (CAP `severity`, `certainty`)	Categorical (e.g. Minor / Unlikely)	tsunami.gov	Assessment by the warning centers
`tsunami` flag	0 / 1	USGS real-time	A tsunami bulletin *may* exist; it does **not** mean that a tsunami occurred

Table 1: Severity-related variables used by the system.

Note on depth in HazEL. The HazEL Swagger defines `eqDepth` as an integer (int32) and does not state its unit. We interpret it as kilometers because it matches the USGS depth (km) for the matched events: in 25 of the 28 NOAA–USGS pairs of the sample (Section 6 of `notebooks/01_source_profiling.ipynb`) the difference is 2 km or less. Being an integer, it loses the decimals that USGS provides.

1.2 Why Monitoring Matters

Tsunamis can reach coasts thousands of kilometers from their source and cause major loss of life and damage, as in the 2011 Tohoku event (M 9.1), which appears in our samples. Monitoring recent earthquakes together with the historical tsunami record and the official bulletins helps an analyst judge quickly how relevant a new earthquake may be. The system supports monitoring

and correlation and does **not** predict tsunamis; the indicator must never be read as a warning, and official warnings come only from the tsunami warning centers.

1.3 Geographic and Temporal Scope

The project covers the Pacific basin, including Alaska and Hawaii, during 2010–2025, with earthquakes of magnitude $M \geq 6$. NOAA HazEL regions 80–89 and a USGS rectangle (latitude -60 to 65, longitude 100 to 300) provide geographic filters, and a Pacific polygon removes the non-Pacific parts of the rectangle (Section 2); the USGS historical sample covers 2010–2011 to meet the workshop's 200-record requirement. The two real-time feeds are profiled in full, and their Pacific events are selected with the same polygon; tsunami.gov covers the Pacific, Alaska, the US and the Caribbean.

Source	Coverage	Fits the scope?
NOAA HazEL	Global	Yes, regions 80–89 in the query, then the Pacific polygon
USGS FDSN	Global	Yes, Pacific rectangle in the query, then the Pacific polygon
USGS real-time feeds	Global	Yes, events selected with the Pacific polygon
tsunami.gov	Pacific, Alaska, US, Caribbean	Yes, messages selected by the coordinates of their earthquake (Pacific polygon)

1.4 Meaning of "Current Situation"

For this system the current situation comprises:

- earthquakes with $M \geq 4.5$ inside the Pacific scope during the last 24 hours and the last 7 days (USGS real-time feed);
- the latest tsunami.gov message of each warning center (NTWC and PTWC), including information statements;
- earthquakes of the last 7 days flagged by USGS with `tsunami = 1`.

$M \geq 4.5$ is the minimum magnitude of the USGS "past month" feed that we use, and these earthquakes feed the recent-events list of the dashboard. In contrast, the statistics, the comparison with the historical record and the indicator use only earthquakes with $M \geq 6$, the threshold of the historical catalog, so that both sides of the comparison follow the same criterion and small earthquakes today do not look unusual only because the history does not contain them.

2 Historical Sources

2.1 NOAA/NCEI HazEL – Global Historical Tsunami Database

Owner	NOAA National Centers for Environmental Information (NCEI)
URL	https://www.ngdc.noaa.gov/hazel/view/swagger
Access method	REST API, JSON
Authentication	None required
Rate limits	Not documented. Team policy: identified User-Agent (`UDistrital-Team8-Tsunami-Monitoring/1.0`), three retries with increasing wait, sequential requests, and caching of every downloaded file.
License and attribution	Public use, no use constraints, according to the dataset metadata. Required citation: National Geophysical Data Center / World Data Service, *NCEI/WDS Global Historical Tsunami Database*, NOAA National Centers for Environmental Information, doi:10.7289/V5PN93H7, accessed 8 November 2026 [1].
Fields used	`id`, `year`, `month`, `day`, `hour`, `minute`, `second`, `latitude`, `longitude`, `causeCode`, `eventValidity`, `eqMagnitude`, `eqDepth`, `maxWaterHeight`, `warningStatusId`, `regionCode`, `numRunups`
Time coverage	The database goes back to 2100 BC; we use 2010–2025 (last event in the sample: 12 December 2025).
Update policy	Curated catalog; its metadata record gives the update frequency as "as needed". Records can be revised or reclassified (for example their validity).

Evidence. Query used to obtain page 1; the unmodified response is stored in `workshop_1/samples/noaa_tsunamis_pacific_2010_2025.json` (page 2 in `noaa_tsunamis_pacific_page2.json`, code tables in `noaa_descriptor_*.json`):

```
GET https://www.ngdc.noaa.gov/hazel/hazard-service/api/v1/tsunamis/events
    ?minYear=2010&maxYear=2025&itemsPerPage=200
    &regionCode=80,81,82,83,84,85,86,87,88,89&page=1
```

A short excerpt of the response is in Appendix A.

Volume and partitioning. The response reports `totalItems` = 221 over two pages (200 and 21 records), about 94 KB of JSON for 2010–2025 and regions 80–89. The bulk load iterates over `page` = 1 to `totalPages` and checks that the number of records equals `totalItems` and that no event id is repeated.

Run-ups. Run-up records (for example 6,429 for a single event) form a separate and much larger table. We do not load them: the indicator works at event level, and the events table already carries a per-tsunami summary (`maxWaterHeight` and `numRunups`). One page of run-ups for event 4682 is kept as evidence (`noaa_runups_event_4682_page1.json`); loading the full table is left as future work.

2.2 USGS FDSN Event Web Service

Owner	U.S. Geological Survey, Earthquake Hazards Program
URL	https://earthquake.usgs.gov/fdsnws/event/1/
Access method	REST API, CSV (also GeoJSON and QuakeML)
Authentication	None required
Rate limits	Maximum of 20,000 events per request (error 400 if exceeded) [2]. No numeric limit on requests per minute is documented; the documentation defines time × area limits that are not enforced yet but may be, and recommends the real-time feeds for automated applications. Team policy: one request per year, sequential, with caching.
License and attribution	U.S. public domain: most USGS information may be used without restriction. Credit is requested, not mandatory [3]; credit line used by the team: "USGS Earthquake Catalog data courtesy of the U.S. Geological Survey".
Fields used	`id`, `time`, `latitude`, `longitude`, `depth`, `mag`, `magType`, `net`, `place`, `status`, `updated`
Time coverage	Catalog of several decades; we load 2010–2025 with $M \geq 6$ inside the Pacific rectangle, then apply the Pacific polygon.
Update policy	Reviewed events are revised later (`updated`, magnitude and location can change). The service offers the `updatedafter` parameter, and deleted events return HTTP 409 unless `includedeleted` is requested; see Section 3.

Evidence. Query used for the historical sample; the unmodified response is stored in `workshop_1/samples/usgs_earthquakes_2010_2011_m6_pacific.csv`:

```
GET https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv
    &starttime=2010-01-01&endtime=2012-01-01&minmagnitude=6
    &minlatitude=-60&maxlatitude=65
    &minlongitude=100&maxlongitude=300
```

The download script compares the number of downloaded records (343) with the result of the `count` method. A short excerpt of the response is in Appendix A.

Volume and partitioning. The `count` method gives 1,930 events for 2010–2025 (about 0.33 MB at roughly 169 bytes per row, estimated from the 343-row sample of 58 KB), as detailed in Table 2.

Year	Events	Year	Events
2010	157	2018	113
2011	186	2019	129
2012	110	2020	95
2013	115	2021	121
2014	130	2022	106
2015	121	2023	124
2016	127	2024	90
2017	86	2025	120

Table 2: USGS events per year ($M \geq 6$, Pacific rectangle), obtained with the `count` method on 8 November 2026. Total: 1,930.

Although these 1,930 events fit under the limit of 20,000 per request, we load one calendar year per request (16 requests). The documentation also defines time × area limits that are not enforced today but could be: a single 2010–2025 query over our rectangle amounts to about 146 million square-degree-days (5,844 days × 25,000 square degrees), above the 100 million cap of the `count` method, while each yearly partition amounts to about 9.15 million. Partitioning also allows retrying a single failed year without repeating the others and makes the load resumable. Each response is stored unchanged in the raw landing zone as `usgs_YYYYMMDD_HHMMSS.csv`.

Pacific polygon. The rectangle is only a coarse query filter: it also contains areas outside the Pacific basin, such as the Caribbean and the Gulf of Mexico (longitudes 270 to 300) and parts of the Indian Ocean (south of Sumatra and Java, and off western Australia). Therefore a Pacific polygon filter is applied afterwards, during the load into the database, and the events it removes are counted in the `ingestion_run` record. The polygon is stored as GeoJSON in `config/pacific_polygon.geojson` and has the 25 vertices of Table 3. It follows the Pacific margins slightly inland (Alaska and British Columbia, the Americas, Australia and East Asia), extends south to latitude -60, and passes along the southern coasts of Java, Bali and Timor, so that the inner Indonesian seas (Java, Flores and Banda) are included while the Sunda trench on the Indian Ocean side is excluded, consistently with HazEL regions 80–89. Because the polygon crosses the antimeridian, its longitudes are stored as a continuous sequence from -255.5 to -66; before the point-in-polygon test, every event longitude $\lambda > 0$ is converted to $\lambda - 360$.

#	Latitude	Longitude	#	Latitude	Longitude
1	66.1°N	168.5°W	14	44.5°S	144°E
2	63.5°N	150°W	15	38°S	142°E
3	61.5°N	140°W	16	25°S	135°E
4	59.5°N	133°W	17	12°S	132°E
5	56°N	129°W	18	9.2°S	124°E
6	50°N	124°W	19	8.95°S	115°E
7	47°N	121°W	20	7.5°S	110°E
8	7.8°N	77.5°W	21	6°S	110°E
9	19.3°S	67.4°W	22	1.5°N	104.5°E
10	53.6°S	71°W	23	17.1°N	108°E
11	56°S	66°W	24	50°N	138.4°E
12	60°S	66°W	25	63°N	154.5°E
13	60°S	144°E			

Table 3: Vertices of the Pacific polygon, in order; the polygon closes from vertex 25 back to vertex 1.

Applied to the samples, the polygon removes 18 of the 343 USGS events (the 2010 Haiti earthquakes in the Caribbean, eight events on the Indian Ocean side of Sumatra, Java and Bali, one event in the Drake Passage, and seven inland earthquakes in South America and northeast Asia, five of them deeper than 500 km) and 7 of the 221 HazEL events. The seven HazEL events are landslide-generated waves in rivers, lakes and reservoirs far from the coast (in China, eastern Russia, the Yukon, British Columbia and Washington); they are outside the scope of a coastal tsunami monitor and are excluded on purpose.

3 Incremental Sources

3.1 USGS Real-Time GeoJSON Feed

Owner	U.S. Geological Survey, Earthquake Hazards Program
URL	https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php
Feed used	Magnitude 4.5+, past 30 days (`summary/4.5_month.geojson`)
Access method	HTTP GET of a static file, GeoJSON
Authentication	None required
Rate limits	No numeric limit is documented; USGS recommends these feeds, rather than the FDSN service, for automated applications [4]. The response carries `Cache-Control: max-age=60`, so polling more than once per minute returns no new data. Team policy: identified User-Agent, at most one request per minute, retries with increasing wait.
License and attribution	U.S. public domain; credit requested [3], same credit line as in Section 2.
Fields used	`id`, `time`, `updated`, `mag`, `magType`, `place`, `status`, `type`, `tsunami`, `net`, and `geometry.coordinates` (longitude, latitude, depth in km)
Coverage	Global, rolling window of the last 30 days, $M \geq 4.5$; Pacific events are selected with the polygon of Section 2.
Update frequency	Regenerated every minute (`Last-Modified` advances each minute; `Expires` is 60 seconds later).
Change detection	Stable event `id`; the `updated` field (epoch milliseconds) changes whenever an event is revised. The HTTP `Last-Modified` header only shows that the file was regenerated, not that an event changed, and no `ETag` is sent.

3.2 tsunami.gov Warning Centers

Owner	NOAA National Weather Service, U.S. Tsunami Warning Centers: National Tsunami Warning Center (NTWC, Palmer, Alaska) and Pacific Tsunami Warning Center (PTWC, Honolulu, Hawaii)
URL (NTWC)	`https://www.tsunami.gov/events/xml/PAAQAtom.xml`
URL (PTWC)	`https://www.tsunami.gov/events/xml/PHEBAtom.xml`
Access method	HTTP GET; Atom feed (XML), each entry links to a CAP 1.2 message (XML) and a text bulletin
Authentication	None required
Rate limits	Not documented. Team policy: identified User-Agent, polling every 5 minutes (messages are rare), retries with increasing wait, and caching of every downloaded file.
License and attribution	Products of the U.S. National Weather Service, a U.S. Government agency, in the public domain; the source is credited as "NOAA/NWS U.S. Tsunami Warning Centers" [5].
Fields used	Atom: entry `id`, `title`, `updated`, `geo:lat`, `geo:long`, CAP link. CAP: `identifier`, `sent`, `msgType`, `event`, `severity`, `urgency`, `certainty`, and the parameters `EventPreliminaryMagnitude`, `EventPreliminaryMagnitudeType`, `EventOriginTime`, `EventDepth` and `EventLatLon`
Coverage	NTWC: coasts of the continental U.S., Alaska and Canada. PTWC: Hawaii, the U.S. Pacific territories and international Pacific and Caribbean coasts. Messages are issued for earthquakes anywhere that may affect these coasts.
Content	Only the latest message of each center; older messages disappear from the feed. Most messages are information statements (for example, CAP `event` = Tsunami Information, `severity` = Minor, `certainty` = Unlikely); warnings are rare.
Change detection	A new message has a new entry `id` and CAP `identifier`; the entry `updated` time and the HTTP `Last-Modified` header change when a message is published. No `ETag` is sent. Because only the latest message is kept, a message issued and replaced between two polls is lost, which limits the polling interval.

3.3 Empirical Update Behavior

Two snapshots were taken on 7 November 2026 at 05:04 and 14:03 UTC (about 9 hours apart). All times in this report are given in UTC, followed by the local time in Bogotá (UTC−5, no daylight saving time); the snapshots correspond to 00:04 and 09:03 in Bogotá.

	USGS feed	tsunami.gov (NTWC, PTWC)
First snapshot (records)	714	1 per feed
Second snapshot (records)	726	1 per feed
New	22	0
Modified	25	0
Removed	10	0

Table 4: Changes between the two snapshots.

In the 9-hour window the USGS feed went from 714 to 726 records: 22 new, 25 modified and 10 removed, while tsunami.gov did not change at all. This reflects how rare messages are, not a fault of the feed: a later check of the feed headers on 9 November showed that the PTWC feed had been updated on 8 November 2026 at 06:35 UTC (01:35 Bogotá), after the snapshots, while the NTWC feed still carried its 5 November message (05:02 UTC, 00:02 Bogotá). All snapshot figures in this section are computed in Section 3.5 of the notebook `notebooks/01_source_profiling.ipynb`.

Removed events. The 10 removed events were not deleted: all of them expired from the rolling 30-day window. The 30-day cutoff of the second snapshot is 8 October 2026 at 14:03 UTC (09:03 Bogotá); the removed events have origin times between 05:06 and 11:36 UTC of that day (00:06 and 06:36 Bogotá), and the oldest event kept in the second snapshot is from 14:11 UTC (09:11 Bogotá).

Modified events. All 25 kept their `id`, and all 25 have a new `updated` value, which is how USGS signals a revision. Of these, 21 were relocated: the epicenter moved between 1.5 and 29 km (median 5.9 km) and the depth changed by up to 49 km; consequently the origin `time`, `place` and `title`, and the location-quality fields `nst`, `dmin` and `rms`, changed too. Six events also changed their magnitude by 0.1 to 0.2 units, always keeping the magnitude type `mb`. The remaining 4 events changed only `updated` or the `felt` count, with no change to the event itself. All 726 events are already `reviewed`, so a reviewed status does not mean that the record is final.

New events. The 22 new events have magnitudes between 4.5 and 5.5 (none with $M \geq 6$); 19 of them lie inside the Pacific polygon. Note that `workshop_1/compare_incremental_samples.py` compares only the `properties` of each event; the notebook also compares `geometry`, which is where the relocations appear.

3.4 Revised or Deleted Records and Watermark Implications

Section 3.3 shows that records change after publication, and that the changes arrive late: the 25 modified events were revised between 1 and 29 days after their origin time (median 18 days; Section 3.5.2 of the notebook). The incremental design therefore cannot rely on origin times or on new identifiers alone. The rules below are applied per source.

USGS real-time feed.

- **Watermark:** the largest `updated` value stored, not `time`. A revision moves `updated` forward but keeps the origin time, so a watermark on `time` would miss all 25 revisions of Section 3.3.
- **Per-event check:** the feed cannot be asked for "changes since", so every poll reads the whole 30-day file. An event is processed only if its `id` is new or its `updated` is greater than the stored value for that `id`.

- **Duplicate key:** (`source`, `id`). Writes are upserts, so a revision updates the existing row instead of creating a new one.
- **Content hash:** a SHA-256 hash of the normalized fields (time, latitude, longitude, depth, magnitude and its type, place, status, tsunami flag). A new version is recorded only when the hash changes; in Section 3.3, 4 of the 25 modified events changed only `updated` or `felt` and would not create a new version.
- **Disappearance:** an event that leaves the feed because its origin time is older than 30 days is expected (all 10 removals in Section 3.3) and its row is kept unchanged. An event that disappears while still younger than 30 days is a deletion candidate: it is confirmed with an FDSN query by event id (HTTP 409 for deleted events) and then marked as deleted, never physically removed.
- **Overlap window:** revisions made after an event has left the 30-day window are not visible in the feed. A daily FDSN query with `updatedafter` set to the watermark minus one day (the overlap) catches them for the $M \geq 6$ Pacific scope.

USGS FDSN catalog. The historical load uses the same duplicate key, so the FDSN and real-time records of one earthquake share one row. Later revisions of historical events reach the database through the daily `updatedafter` query described above.

tsunami.gov.

- **Revisions:** no published message was revised in the snapshots; a correction or update is issued as a new message with a new identifier.
- **Watermark:** the entry `updated` time and the CAP `identifier` of the last stored message of each center (NTWC and PTWC).
- **Duplicate key:** the CAP `identifier`; messages are insert-only.
- **Cheap polling:** the stored `Last-Modified` value is sent back as `If-Modified-Since`, so an unchanged feed is not processed again; whether the server answers 304 must still be tested in Workshop 3.
- **Limitation:** each feed keeps only its latest message, so a message that is replaced between two polls is lost. Polling every 5 minutes makes this unlikely, and the limitation is stated in the dashboard.

NOAA HazEL. HazEL records have no update timestamp, yet they can be reclassified (for example their `eventValidity`), so no watermark is possible. HazEL is not an incremental source: each historical run downloads the 221 records of regions 80–89 again (two requests) and compares them by content hash; changed records are updated and unchanged ones are skipped.

Failed runs. For every source the watermark is written in the same database transaction as the records it covers, so it advances only after a successful commit. Because all writes are upserts by key, re-running a failed or partial run processes the same records again without creating duplicates (see Section 6).

4 Source Profiling and Data Quality

4.1 Profiling Summary

Every sample was profiled in `notebooks/01_source_profiling.ipynb` (Sections 3 to 6): null percentage, type and range of every field, duplicated identifiers, timestamp formats and time zones, coordinate validity and units. Table 5 summarizes the result. No source has duplicated identifiers inside a snapshot, and no coordinate is outside the WGS84 ranges or at $(0, 0)$; the only invalid locations are 9 HazEL events without coordinates. All timestamps are in UTC, but in

four different formats: HazEL splits the date into six columns (`year` to `second`, with fractional seconds and with `hour` and `minute` sometimes missing), the FDSN CSV uses ISO 8601 with `Z`, the real-time feed uses epoch milliseconds, and tsunami.gov uses ISO 8601 (with `Z` in the Atom feed and the offset `-00:00` in CAP). Depth is given in kilometers by USGS, as an integer without unit by HazEL, and as text with its unit (`473 kilometers`) by the CAP messages. Magnitude comes with its type in USGS (6 types in each sample: `mww`, `mwc`, `mb`, `mwb`, `mw`, `ms` in the catalog; `mb` for 609 of 726 feed events), without type in HazEL, and as a preliminary value with its type (`Ml`, `Mwp`) in tsunami.gov.

Source	Records	Dupl. ids	Notes
NOAA HazEL	221	0	56 columns; 2010-01-02 to 2025-12-12. Impact fields (deaths, damage, houses) are 63–99.5% null; `warningStatusId` 76.5% null; `eqMagnitude` and `eqDepth` 15.8% null (tsunamis without an earthquake cause); 9 events without coordinates.
USGS FDSN	343	0	24 columns; 2010–2011, M 6.0–9.1, depth 2.8–644.6 km. `magError` 100% null; `dmin` and `horizontalError` 99.4% null; `depthError` 80.8% null. All events `reviewed`.
USGS real-time	726	0	32 columns; 8 October to 7 November 2026, M 4.5–7.7, depth 1–643 km. `tz` 100% null; `alert` 89.8% null; `felt`, `cdi` 82.4% null. All events `reviewed`; 6 with `tsunami` = 1.
tsunami.gov	2	0	One message per center (NTWC and PTWC), which is the full content of each feed, so the 200-record minimum cannot apply. Both are information statements (`Minor` / `Unlikely`).

Table 5: Profiling summary of the four samples (second snapshot for the incremental sources).

4.2 Data-Quality Problems

Seven problems were found in the samples; each one is computed in Section 5 of the notebook, except problem 6, which is shown in Section 6.1.1 of the notebook while matching records (Section 4.3).

#	Source	Problem	Example record
1	NOAA HazEL	Official warning status missing: 169 of 221 events (76%) have no `warningStatusId`.	Event 4522 (Solomon Islands, 3 January 2010): definite tsunami (validity 4) with no warning status.
2	NOAA HazEL	Erroneous or doubtful events: 9 events with `eventValidity` ≤ 2 (8 questionable, 1 erroneous entry).	Event 5407 (South Island, New Zealand, 21 February 2011): questionable tsunami (validity 2).
3	NOAA HazEL	Time of day or coordinates missing: 23 events have only a date, and 9 events have no coordinates.	Event 4762 (Lake 513, Hualcan glacier, Peru, 13 April 2010): no hour or minute. Event 5739 (Cheju Strait, 2 January 2010): no latitude or longitude.
4	USGS FDSN	Columns that are always or almost always empty: `magError` is 100% null, `dmin` and `horizontalError` 99.4%.	Event `official20110311054624120_30` (Tohoku, 11 March 2011): empty `magError`, `horizontalError` and `dmin`.
5	tsunami.gov	Location as free text and depth as text with its unit; the location text uses miles.	NTWC message `PAAQ-1-to0j1g`: "75 miles SW of Eureka, California"; CAP `EventDepth` = "13 kilometers".
6	tsunami.gov	Wrong origin date in the CAP message: the origin time is one day earlier than the real earthquake and than the message itself.	PTWC message `PHEB-1-26297050`, sent 25 October 2026 at 02:27 UTC, gives `EventOriginTime` 24 October 02:09 UTC; USGS `us6000u4r9` places the earthquake on 25 October at 02:09:24 UTC.
7	USGS real-time	The `tsunami` flag can be misread: it means that a bulletin may exist, not that a tsunami happened (6 of 726 events).	Event `us6000u4r9` (Bonin Islands, M 6.5, depth 492 km): `tsunami` = 1, while the PTWC message for it says `Minor` / `Unlikely`.

Table 6: Data-quality problems found in the samples.

In addition, 36 of the 221 HazEL events have a non-earthquake cause (Section 1), so their magnitude and depth are empty by nature and must not be treated as missing values.

4.3 Cross-Source Overlap

For this exploration two records are taken as the same event when their origin times differ by at most 15 minutes and their epicenters are at most 100 km apart (Section 6 of the notebook); the final rule will be justified in Workshop 3. With this rule, all 28 earthquake-caused HazEL tsunamis of 2010–2011 were matched to a USGS FDSN event, with a median time difference of 0 s, a median distance of 0 km and a median magnitude difference of 0; the HazEL depth is within 2 km of the USGS depth in 25 of the 28 pairs. Between the two incremental sources the rule matched only the NTWC message; the PTWC message was matched only after widening the time window to two days (Section 6.1.1 of the notebook), which revealed the date error of problem 6: the matching USGS event is 24.01 hours later than the CAP origin time. Table 7 lists six pairs (Δt is the time of record B minus the time of record A).

#	Record A / Record B	Δt	Distance	Magnitude A / B	What differs
1	tsunami.gov `PAAQ-1-to0j1g` / USGS feed `nc75462637`	−1.3 s	8.7 km	4.5 Ml / 4.96 mw	Magnitude value and type, depth (13 vs 10 km), location text
2	HazEL 5413 / USGS `official2011 0311054624120_30`	+0.02 s	0.0 km	9.1 / 9.1 mww	Identifier format; HazEL time rounded to 0.1 s and magnitude without type
3	HazEL 4682 / USGS `official2010 0227063411530_30`	+0.03 s	0.0 km	8.8 / 8.8 mww	Depth 22 vs 22.9 km (HazEL integer)
4	HazEL 5406 / USGS `usp000huc2`	+0.21 s	15.9 km	6.8 / 6.9 mww	Epicenter, depth (28 vs 26 km) and magnitude
5	HazEL 5931 / USGS `usp000hvkh`	+0.68 s	0.0 km	6.3 / 6.5 mwc	Magnitude (0.2 lower in HazEL); HazEL time without decimals
6	tsunami.gov `PHEB-1-26297050` / USGS feed `us6000u4r9`	+1 day +24 s	35.8 km	7.0 Mwp / 6.5 mww	Origin date (problem 6), preliminary magnitude 0.5 higher, depth (473 vs 492 km)

Table 7: Candidate pairs of records that describe the same real-world event.

Two lessons follow for the matching rule: magnitude must not be required to be equal (differences of 0.1 to 0.5 and different magnitude types are normal), and tsunami.gov times cannot be trusted alone, so a message should also be linked through its location and magnitude when no USGS event lies within the time window.

5 Canonical Mapping

5.1 Field Mapping to the Canonical `event`

Canonical field	NOAA HazEL	USGS FDSN / feed	tsunami.gov (CAP)	Rule
`event_id`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`source`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`event_type`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`timestamp` (UTC)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`latitude`	[TODO:]	[TODO:]	[TODO:]	WGS84
`longitude`	[TODO:]	[TODO:]	[TODO:]	WGS84, −180 to 180
`severity`	[TODO:]	[TODO:]	[TODO:]	[TODO: value and unit/type]
`raw_reference`	[TODO:]	[TODO:]	[TODO:]	id or URL plus content hash

5.2 Domain Tables

[TODO: Proposed tables (for example `earthquake`, `tsunami_event`, `tsunami_alert`) with their attributes.]

5.3 Unit Conversions and Rules

[TODO: Timestamps (UTC), coordinates (WGS84), longitude normalization, depth units, magnitude types and severity scale.]

5.4 Draft ER Diagram



Figure 1: First draft of the entity-relationship diagram.

6 Ingestion Contract Draft

Source	Historical strategy	Incremental strategy	Duplicate key	Overlap window
NOAA HazEL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS FDSN	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS real-time feed	–	[TODO:]	[TODO:]	[TODO:]
tsunami.gov	–	[TODO:]	[TODO:]	[TODO:]

6.1 Correctness After a Failed Run

[TODO: The watermark advances only after a successful commit; idempotent writes; what happens to a partially loaded batch.]

7 Project Definition

7.1 Project Overview

[TODO: Title, motivation, goal (data-engineering and analytical), scope, explicit exclusions and constraints.]

7.2 Functional User

[TODO: Role, profile, decisions supported, and why the dashboard is useful.]

7.3 User Stories

ID	Type	Story	MoSCoW
US1	Monitoring	As a [TODO: user], I want to [TODO: action] so that [TODO: benefit].	[TODO:]
US2	Monitoring	[TODO:]	[TODO:]
US3	Analysis	[TODO:]	[TODO:]
US4	Analysis	[TODO:]	[TODO:]
US5	Trust	[TODO:]	[TODO:]
US6	Trust	[TODO:]	[TODO:]

7.4 Indicator Proposal

[TODO: Name, inputs (magnitude, depth, location, historical tsunami association, official alert), rule, output scale, how it will be validated against historical events. State clearly that it is a monitoring aid and not a prediction, and list its limitations.]

7.5 Risks and Mitigation

Risk	Impact	Mitigation
Source outage	[TODO:]	[TODO:]
Rate limits	[TODO:]	[TODO:]
Schema change	[TODO:]	[TODO:]
License and attribution	[TODO:]	[TODO:]
Ambiguous identifiers	[TODO:]	[TODO:]

7.6 Architecture Overview

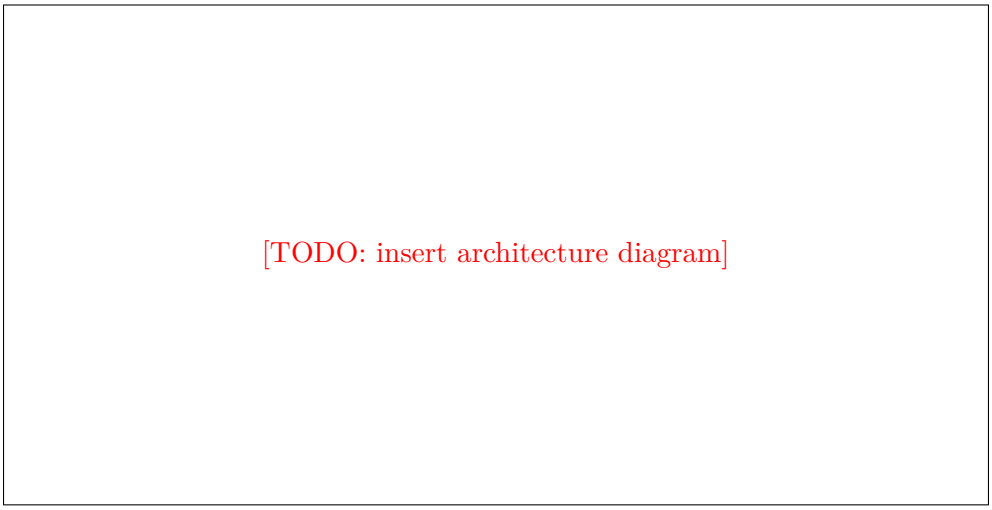


Figure 2: High-level architecture adapted to tsunami monitoring.

Conclusions and Open Questions

[TODO: Main decisions taken in this workshop and what remains open for Workshop 2.]

References

[1] National Geophysical Data Center / World Data Service. *NCEI/WDS Global Historical Tsunami Database*. NOAA National Centers for Environmental Information. doi:10.7289/V5PN93H7. Accessed through the HazEL API, https://www.ngdc.noaa.gov/hazel/view/swagger, 8 November 2026.

[2] U.S. Geological Survey. *API Documentation – Earthquake Catalog*. https://earthquake.usgs.gov/fdsnws/event/1/. Accessed 8 November 2026.

[3] U.S. Geological Survey. *Acknowledging or Crediting USGS*. https://www.usgs.gov/information-policies-and-instructions/acknowledging-or-crediting-usgs. Accessed 8 November 2026.

[4] U.S. Geological Survey. *Real-time Notifications, Feeds, and Web Services (GeoJSON summary feed)*. https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php. Accessed 7 and 9 November 2026.

[5] NOAA National Weather Service, Tsunami Warning Centers. *Tsunami event feeds*. https://www.tsunami.gov/. Accessed 7 and 9 November 2026.

A Real Response Snippets

Short excerpts of the unmodified responses stored in `workshop_1/samples/`. Each excerpt shows one record; "..." marks fields or lines left out for brevity.

A.1 NOAA HazEL – tsunami events

File `noaa_tsunamis_pacific_2010_2025.json`, record of the 2011 Tohoku tsunami, and the pagination fields of the response:

```
{"items": [ ...
  {"id": 5413, "year": 2011, "month": 3, "day": 11, "hour": 5, "minute": 46,
   "latitude": 38.297, "longitude": 142.373, "causeCode": 1,
   "eventValidity": 4, "eqMagnitude": 9.1, "eqDepth": 29,
   "maxWaterHeight": 39.26, "warningStatusId": null, "regionCode": 85,
   "numRunups": 6429, "country": "JAPAN", "locationName": "HONSHU ISLAND", ...},
  ... ],
 "page": 1, "totalPages": 2, "itemsPerPage": 200, "totalItems": 221}
```

A.2 USGS FDSN – earthquake catalog

File `usgs_earthquakes_2010_2011_m6_pacific.csv`, header and the row of the same earthquake:

```
time,latitude,longitude,depth,mag,magType,nst,gap,dmin,rms,net,id,updated,place,type,
    horizontalError,depthError,magError,magNst,status,locationSource,magSource
2011-03-11T05:46:24.120Z,38.297,142.373,29,9.1,mww,541,9.5,,1.16,official,
    official20110311054624120_30,2026-10-16T10:38:09.160Z,"2011 Great Tohoku Earthquake, Japan",
    earthquake,,,,,reviewed,us,official
```

The two records describe the same event: same origin minute, epicenter and magnitude, and the HazEL `eqDepth` (29) equals the USGS depth in km.

A.3 USGS real-time feed

File `usgs_realtime_20261107_050434.geojson`, feed metadata and first feature (times in epoch milliseconds, coordinates as longitude, latitude, depth in km):


```
{"type": "FeatureCollection",
 "metadata": {"generated": 1794027872000,
   "url": "https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/4.5_month.geojson",
   "title": "USGS Magnitude 4.5+ Earthquakes, Past Month", "status": 200, "count": 714, ...},
 "features": [
  {"type": "Feature", "id": "us7000twmr",
   "properties": {"mag": 4.8, "place": "36 km WNW of Poso, Indonesia",
     "time": 1793992021269, "updated": 1794002237040, "tsunami": 0,
     "magType": "mb", "status": "reviewed", "type": "earthquake", ...},
   "geometry": {"type": "Point", "coordinates": [120.4589, -1.2393, 10]}},
  ... ]}
```


A.4 tsunami.gov – PTWC Atom feed

File `ptwc_20261107_050434.xml`, the single entry of the feed (an information statement); the linked CAP message is stored in `ptwc_cap1_20261107_050434.xml`:


```
<entry>
  <title>BONIN ISLANDS JAPAN REGION</title><updated>2026-10-25T02:27:01Z</updated>
  <geo:lat>28.</geo:lat>
  <geo:long>139.9</geo:long>
  <summary type="xhtml"> ... <strong>Category:</strong> Information<br/>
    <strong>Bulletin Issue Time: </strong> 2026.10.25 02:27:01 UTC <br/>
    <strong>Preliminary Magnitude: </strong>7.0(Mwp)<br/> ... </summary>
  <id>urn:uuid:ba64bc82-07f7-45fb-ab6c-25420fca6851</id>
  <link rel="related" title="CapXML document" type="application/cap+xml"
       href="https://www.tsunami.gov/events/PHEB/2026/10/24/26297050/1/WEGM42/PHEBCAP.xml" />
</entry>
```